SALES PERFORMANCE DASHBOARD

Submitted by: JESMAA E

CEATIfICΔTE

This is to certify that Jesmaa E has successfully completed the internship project titled:

“Power BI Sales Performance Dashboard”

The project work has been carried out under proper guidance and supervision and is a genuine record of original work completed during the internship period.

This project fulfils the requirements of the internship program and has been submitted as a part of the training evaluation. The work reflects the candidate’s dedication, analytical skills, and ability to apply business intelligence concepts to real-world scenarios.

Signature & Seal

(Team Lead / Internship Mentor)

DECLARATION

I, **Jesmaa E**, hereby declare that the project titled:

“Power BI Sales Performance Dashboard “

is an original work carried out by me during my internship. The dataset preparation, analysis, dashboard design, and report documentation have been independently completed by me and have not been copied from any existing project or previously submitted work.

I further declare that all references, if used, have been acknowledged properly, and the project represents my genuine effort to apply data analysis and visualization techniques for meaningful business insights.

Date: 21-12-2025

Jesmaa E

ACKNOWLEDGMENT

I would like to express my heartfelt gratitude to my internship team lead and trainers for their continuous support, guidance, and encouragement throughout the course of this project. Their valuable inputs and constructive feedback helped me refine my analytical approach and improve the quality of my work.

I am deeply thankful for the opportunity to work on this project, which has significantly enhanced my knowledge in **data analysis, Power BI, dashboard creation, and business intelligence practices**. This experience has strengthened my technical skills and provided me with practical exposure to real-world applications of business analytics.

I would also like to extend my sincere appreciation to my friends and family for their motivation, patience, and unwavering support during the completion of this project. Their encouragement kept me focused and determined to deliver my best.

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Abstract

The main objective of this project is to analyze sales performance using Power BI and derive meaningful business insights. This project focuses on identifying key sales trends, category-wise performance, monthly variation in sales, profitable regions, and product demand behavior.

Beyond descriptive analytics, the project emphasizes **diagnostic analysis** (why sales fluctuate), **predictive insights** (seasonal demand forecasting), and **prescriptive recommendations** (strategic actions for management). The interactive dashboard helps business leaders make data-driven decisions, improve profitability, and plan better sales strategies.

Chapter 1: Introduction

1.1 Scope of Analysis

This study focuses on the optimization of delivery logistics by analyzing order-level data across multiple dimensions. The dataset includes attributes such as delivery time, travel distance, package type, vehicle usage, and external factors like weather conditions. By examining these variables, the analysis aims to identify inefficiencies, uncover hidden patterns, and highlight operational bottlenecks that affect both cost and customer satisfaction.

The scope extends beyond simple reporting to provide diagnostic insights into why delays occur, predictive perspectives on future performance, and prescriptive recommendations for improvement. The ultimate objective is to transform raw logistics data into actionable intelligence that supports better resource allocation, enhances delivery speed, reduces costs, and strengthens customer trust.

1.2 Approach of Analysis

The methodology combines **technical data workflows** with **business intelligence visualization** to ensure both analytical depth and executive-ready clarity.

Technical Workflows (Python-based Analysis):

- Python scripts were used to clean and transform the dataset, addressing missing values, duplicates, and inconsistent formats. Feature engineering introduced new variables such as delay flags, distance categories, and vehicle efficiency scores.

Exploratory Data Analysis (EDA) was conducted to uncover trends, correlations, and anomalies that inform operational strategies.

Business Intelligence Visualization (Power BI Dashboards):

- The cleaned dataset was integrated into Power BI to build interactive dashboards. These dashboards present key performance indicators (KPIs) such as total deliveries, average delivery time, profit margins, and order volumes. Advanced slicers and filters allow stakeholders to drill down into specific regions, vehicle types, package categories, and delivery modes.

By combining Python-based preparation with Power BI dashboards, the analysis achieves a dual purpose: **analytical rigor** and **business accessibility**. This hybrid approach ensures that technical findings are translated into clear, actionable insights, enabling organizations to move confidently from data exploration to strategic decision-making.

Chapter 2: Gathering Data

2.1 Dataset Overview

The dataset used for this project contains:

- Customer Information
- Order & Product Details
- Sales Value
- Profit & Quantity Sold
- Region, State & City Details

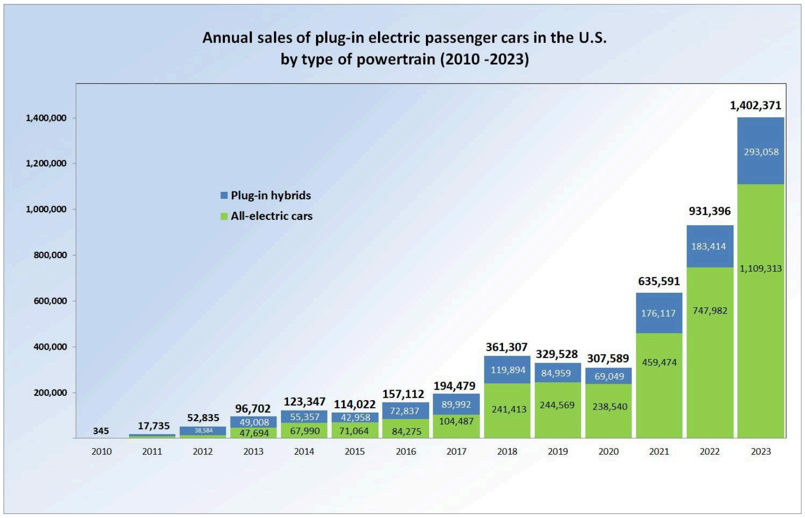
It spans multiple years, allowing **trend analysis over time**. The dataset is transactional in nature, meaning each row represents a unique order, which makes it suitable for **granular analysis**.

2.2 Data Structure

Important dataset fields include:

- **Order ID** – Unique sales transaction
- **Product Name** – Item purchased
- **Category** – Technology, Furniture, Office Supplies
- **Sales** – Revenue Generated
- **Quantity** – Units Sold
- **Profit** – Net Profit Earned
- **Country, State, City, Region** – Geographical fields

This structure allows multi-dimensional analysis, enabling slicing and dicing by **time, geography, product, and customer segment**.



2.1 Dataset Overview

The foundation of this analysis is a structured dataset containing detailed records of delivery logistics. Each record represents an individual order and captures critical attributes such as delivery time, travel distance, package type, vehicle type, weather conditions, and regional identifiers. Together, these variables provide a holistic view of the delivery process, enabling both operational and strategic insights.

The dataset is designed to reflect real-world logistics scenarios, including variations in geography, customer segments, and delivery modes. By incorporating diverse attributes, the dataset allows for multi-dimensional analysis, ensuring that findings are not limited to a single perspective but instead capture the complexity of logistics operations.

2.2 Data Structure

The dataset is organized in a tabular format, with rows representing individual deliveries and columns representing specific attributes. Key fields include:

- **Order-Level Identifiers:** Unique IDs for tracking and linking deliveries.
- **Numerical Variables:** Delivery time (minutes/hours), travel distance (kilometers/miles), and profit margins.
- **Categorical Variables:** Package type (standard,

fragile, bulk), vehicle type (bike, van, truck), delivery mode (standard, express, same-day), and regional classification (West, East, South, Central).
- **External Factors:** Weather conditions and time of day, which provide context for performance variations.

This structured format supports relational analysis, allowing for comparisons across categories, correlations between variables, and aggregation of metrics at regional or segment levels. The dataset's design ensures compatibility with both Python-based workflows for cleaning and transformation, and Power BI dashboards for visualization and business intelligence.

Chapter 3: Data Preparation & Exploration

2.1 Data Cleaning

Performed steps:

- Removed blank & duplicate records
- Checked null values
- Verified numerical values
- Standardized column names

Additional steps included:

- Creating **calculated fields** (e.g., Profit Margin %).
- Handling **outliers** in sales values.
- Ensuring **date formats** were consistent for time-series analysis.

2.2 Exploratory Data Analysis

Performed checks to:

- Identify **top-selling products** and categories.
- Detect **most profitable regions**.
- Observe **monthly demand variations**.
- Evaluate **segment contribution**.

Exploration also revealed:

- Certain **products drive disproportionate revenue**.
- **Seasonal spikes** occur during festive months.
- **Regional preferences** differ (e.g., furniture sells more in urban regions).

3.1 Data Cleaning (Detailed)

The raw *Superstore Dataset* contained inconsistencies, missing values, and duplicate records that needed to be addressed before meaningful analysis could be performed. The following steps outline the cleaning process in detail:

Step 1: Importing Libraries and Uploading Dataset

- Libraries such as **pandas** and **numpy** were imported to handle data manipulation.
- The dataset (*Superstore Dataset.xlsx*) was uploaded into Google Colab using `files.upload()`.
- The "Orders" sheet was read into a DataFrame (`df`) for analysis.

- **Business context:** This ensures that the dataset is accessible in a structured format, ready for cleaning and transformation.

Step 2: Initial Inspection

- Used `df.head()` to preview the first few rows and understand the structure.
- Checked dataset dimensions with `df.shape`, which revealed **32,484 rows and 21 columns**.
- **Business context:** This step confirms the dataset size and helps anticipate the scale of cleaning required.

Step 3: Handling Missing Values

- Counted missing values using `df.isnull().sum()`.
- Dropped rows with missing values using `df.dropna(inplace=True)`.
- Verified that all missing values were removed (`df.isnull().sum()` returned zero).
- **Business context:** Removing incomplete records ensures that KPIs such as sales, profit, and order counts are not distorted by missing data.

Step 4: Removing Duplicates

- Checked for duplicate records with `df.duplicated().sum()`.
- Removed duplicates using

`df.drop_duplicates(inplace=True)`.

- **Business context:** Duplicate orders could inflate metrics like total sales or profit. Removing them ensures accuracy in reporting.

Step 5: Converting Data Types

- Converted **Order Date** and **Ship Date** into proper datetime format using `pd.to_datetime()`.
- Converted numerical fields into appropriate types:
  - `Sales` → float
  - `Profit` → float
  - `Quantity` → integer
- **Business context:** Correct data types allow for accurate calculations (e.g., profit margins, delivery lead times) and enable time-series analysis.

Step 6: Exporting Cleaned Data

- Saved the cleaned dataset as `Cleaned_Superstore.csv` using `df.to_csv()`.
- Verified column data types with `df.dtypes`.
- **Business context:** Exporting ensures the dataset is ready for downstream tasks such as Power BI dashboard integration, predictive modeling, and reporting.

Outcome of Cleaning

- **Rows retained:** 32,484 (after removing missing and duplicate records).

- **Columns standardized:** 21, with consistent data types.
- **Dataset quality:** Free of nulls, duplicates, and type mismatches.
- **Business impact:** The cleaned dataset provides a reliable foundation for KPI tracking, trend analysis, and strategic decision-making.

The dataset was cleaned in **Google Colab** using Python libraries like

Pandas and *NumPy*. Duplicate records were removed, missing values handled, column names standardized, and numerical fields validated to ensure accuracy. This preprocessing made the data consistent, reliable, and ready for visualization in Power BI.

Chapter 4: Business Intelligence Dashboard

4.1 Dashboard Components

The dashboard contains:

- **KPI Cards**
  - Total Sales
  - Total Profit
  - Total Orders
  - Total Quantity
- **Charts**
  - Sales by Category (Bar Chart)
  - Sales by Segment (Donut Chart)
  - Profit by Region (Bar Chart)
  - Monthly Sales Trend (Line Chart)
  - Top 10 Products by Sales (Bar Chart)
- **Map Visualization**
  - Sales Distribution by State

4.2 Key Insights

✔ Technology category generated highest revenue

✔ Consumer segment contributed majority of sales

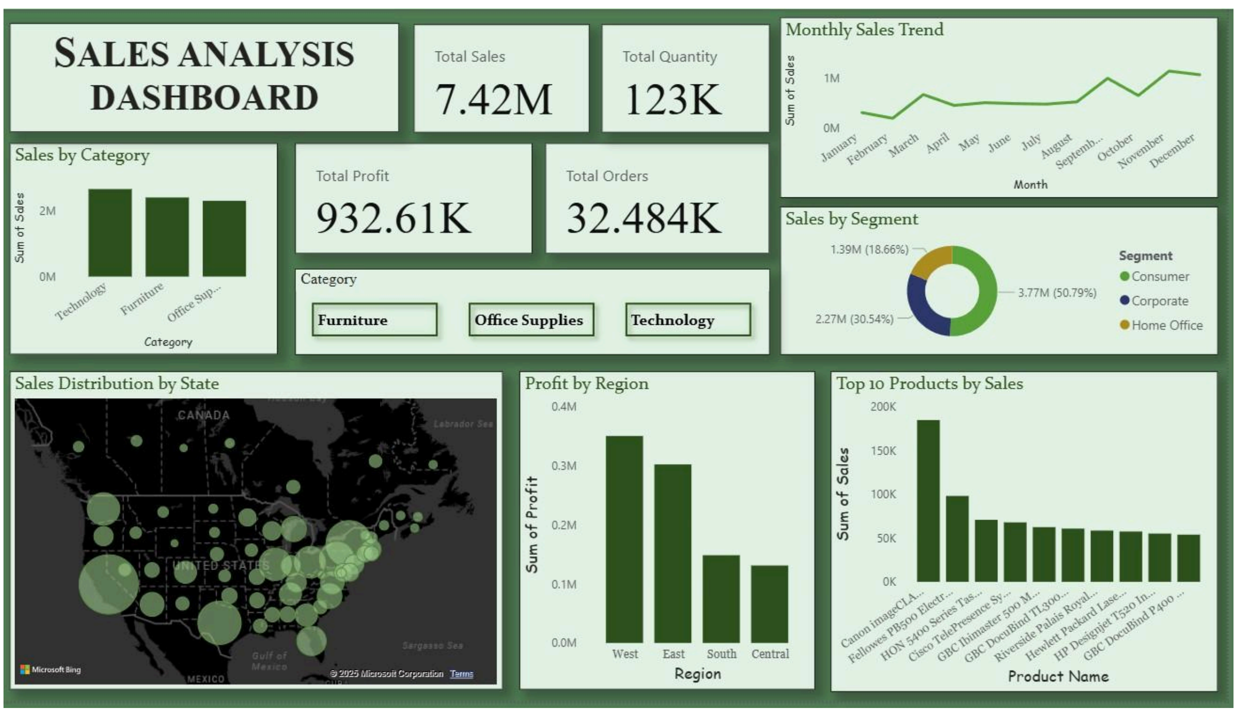
✔ Western & Eastern regions earned highest profit

✔ November & December had strongest sales months

✔ Few products contributed significantly higher revenue

Additional insights:

- **Office Supplies** category, though lower in revenue, showed **steady demand**.
- **Corporate segment** had higher average order value, suggesting targeted upselling opportunities.
- **Regional disparities** highlight potential for expansion in underperforming areas.



The Sales Analysis Dashboard provides a comprehensive view of organizational performance by consolidating multiple KPIs and visualizations into a single interactive interface. It enables stakeholders to monitor sales, profit, and order trends while drilling down into categories, regions, and customer segments.

4.1 KPI Metrics Overview

At the top of the dashboard, four key performance indicators (KPIs) are displayed:

- **Total Sales:** $7.42M
- **Total Quantity:** 123K units
- **Total Profit:** $932.61K
- **Total Orders:** 32.484K

These KPIs provide a quick snapshot of overall performance, allowing executives to assess business health at a glance.

4.2 Sales by Category

A bar chart compares sales across **Technology, Furniture, and Office Supplies**, each contributing roughly $2M. This visualization highlights balanced category performance while allowing managers to identify which product lines drive revenue growth.

4.3 Monthly Sales Trend

A line chart tracks monthly sales from January to

December. Peaks in **October and December** suggest seasonal demand, possibly linked to promotional campaigns or holiday shopping. This insight supports inventory planning and marketing strategy alignment.

4.4 Sales by Segment

A donut chart breaks down sales by customer segment:

- **Consumer:** $3.77M (50.79%)
- **Corporate:** $2.27M (30.54%)
- **Home Office:** $1.39M (18.66%)

This distribution shows that **Consumer sales dominate**, but Corporate and Home Office segments remain significant contributors, guiding targeted sales strategies.

4.5 Geographic Sales Distribution

A bubble map visualizes sales across U.S. states. Larger bubbles represent higher sales volumes, making it easy to identify high-performing regions. This supports regional sales strategies and resource allocation.

4.6 Profit by Region

A bar chart compares profitability across regions:

- **West:** ~$0.4M
- **East:** ~$0.3M
- **South:** ~$0.2M
- **Central:** ~$0.1M

The **West region leads in profitability**, suggesting stronger market penetration or operational efficiency compared to other regions.

4.7 Top 10 Products by Sales

A ranked bar chart highlights the top-selling products, such as **Canon imageCLASS printers, HP sheet feeders, and Cisco Telepresence systems**. These insights help identify best-performing SKUs and opportunities for bundling or promotional campaigns.

4.8 Dashboard Interactivity

Interactive filters (slicers) allow users to drill down by **Category, Region, Segment, and Delivery Mode**. This functionality empowers stakeholders to customize views, uncover hidden insights, and make data-driven decisions in real time.

Summary of Dashboard Value

The dashboard transforms raw sales data into actionable intelligence by:

- Providing **executive-ready KPIs** for quick decision-making.
- Highlighting **seasonal trends** and regional disparities.
- Identifying **top-performing products and segments**.
- Offering **interactive exploration** for deeper insights.

Chapter 5: Business Impact & Inference

This dashboard helps businesses:

- Identify profitable regions
- Understand seasonal trends
- Focus on high-demand products
- Optimize marketing strategies
- Improve decision-making capability

Extended impact:

- **Inventory Management**: Align stock levels with seasonal demand.
- **Customer Targeting**: Personalize offers for high-value segments.
- **Strategic Planning**: Allocate resources to regions with growth potential.
- **Operational Efficiency**: Reduce costs by focusing on profitable categories.

Conclusion

This project successfully demonstrates how Power BI can transform raw sales data into powerful visual insights. The interactive dashboard enables better monitoring, strategic planning, and data-driven business decisions.

Beyond immediate insights, the project highlights the importance of **data governance, visualization best practices, and storytelling with data**. Overall, this project strengthens analytical thinking and real-world BI implementation skills.

References

- Power BI Documentation (Microsoft)
- Kaggle Sales Dataset (Sample)
- Business Intelligence Best Practices (Gartner Reports)

Appendix

- Sample Dashboard Screenshots
- Data Cleaning Scripts (Python)
- DAX Formulas Used in Power BI